Chapter 14, Problem 14P

Problem

Draw the Bode plots for

$$\mathbf{H}(\omega) = \frac{250(j\omega + 1)}{j\omega(-\omega^2 + 10\,j\omega + 25)}$$

Step-by-step solution

Step 1 of 9

Consider the following transfer function:

$$\mathbf{H}(\omega) = \frac{250(j\omega+1)}{j\omega(-\omega^2+10j\omega+25)}$$

Write the transfer function in the standard form.

$$\begin{aligned}\mathbf{H}(\omega) &= \frac{250(j\omega+1)}{j\omega(-\omega^2+10j\omega+25)} \\ &= \frac{250(j\omega+1)}{25j\omega\left(\frac{(j\omega)^2}{25}+\frac{10j\omega}{25}+1\right)} \\ &= \frac{10(j\omega+1)}{j\omega\left(\frac{(j\omega)^2}{25}+\frac{10j\omega}{25}+1\right)} \\ &= 10\frac{1+j\omega}{j\omega\left(1+\left(\frac{2}{5}\right)j\omega+\left(\frac{j\omega}{5}\right)^2\right)}\end{aligned}$$

$$\mathbf{H}(\omega) = 10\frac{1+j\omega}{j\omega\left(1+\frac{j\omega}{5}\right)^2} \quad \text{...... (1)}$$

Step 2 of 9

Write the magnitude expression from equation (1).

$$\begin{aligned} G_{dB} &= 20\log(10)+20\log|1+j\omega|-20\log|j\omega|-20\log\left|1+\frac{j\omega}{5}\right|^2 \\ &= 20+20\log|1+j\omega|-20\log|j\omega|-40\log\left|1+\frac{j\omega}{5}\right| \end{aligned}$$

Determine the magnitude value $\omega = 0.1\ \text{rad/s}$.

$$\begin{aligned} G_{dB} &= 20+20\log|1+j0.1|-20\log|j0.1|-40\log\left|1+j\frac{0.1}{5}\right| \\ &= 20+20\log|1+j0.1|-20\log|j0.1|-40\log|1+j0.02| \\ &= 40\,\text{dB} \end{aligned}$$

Determine the magnitude value $\omega = 1\ \text{rad/s}$.

$$\begin{aligned} G_{dB} &= 20+20\log|1+j1|-20\log|j1|-40\log\left|1+j\frac{1}{5}\right| \\ &= 20+20\log|1+j1|-20\log|j1|-40\log|1+j0.2| \\ &= 22.67\,\text{dB} \end{aligned}$$

Determine the magnitude value $\omega = 5\ \text{rad/s}$.

$$\begin{aligned} G_{dB} &= 20+20\log|1+j5|-20\log|j5|-40\log\left|1+j\frac{5}{5}\right| \\ &= 20+20\log|1+j5|-20\log|j5|-40\log|1+j1| \\ &= 14.15\,\text{dB} \end{aligned}$$

Step 3 of 9

There are three corner frequencies at $\omega = 0,\ 1,\ 5\ \text{rad/s}$.

For the pole with corner frequency at $\omega = 0\ \text{rad/s}$, the slope of the magnitude plot is $-20\ \text{dB/decade}$.

For the zero with corner frequency at $\omega = 1\ \text{rad/s}$, the slope of the magnitude plot is $20\ \text{dB/decade}$.

For the pole with corner frequency at $\omega = 5\ \text{rad/s}$, the slope of the magnitude plot is $-40\ \text{dB/decade}$. Since, there are two poles at same instant.

Step 4 of 9

Draw the asymptotic magnitude plot.

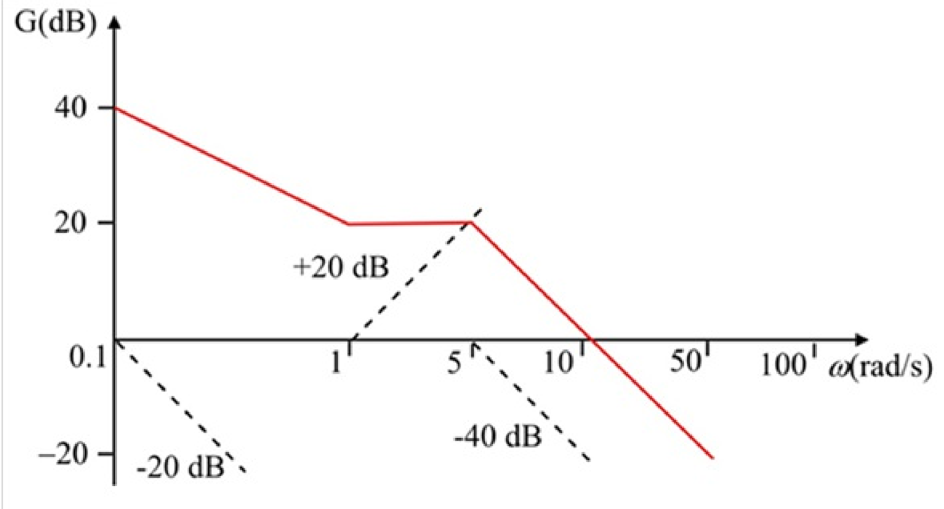

Figure 1

Step 5 of 9

Write the phase expression from equation (1).

$$\phi = \tan^{-1}\left(\frac{\omega}{1}\right) - \tan^{-1}\left(\frac{\omega}{0}\right) - 2\tan^{-1}\left(\frac{\frac{\omega}{5}}{1}\right)$$

$$= \tan^{-1}(\omega) - 90^\circ - 2\tan^{-1}\left(\frac{\omega}{5}\right)$$

$$\phi = -90^\circ + \tan^{-1}(\omega) - 2\tan^{-1}\left(\frac{\omega}{5}\right) \quad \text{...... (2)}$$

Step 6 of 9

Determine the phase angle at $\omega = 0.01\ \text{rad/s}$.

$$\phi = -90° + \tan^{-1}(0.01) - 2\tan^{-1}\left(\frac{0.01}{5}\right)$$
$$= -89.66°$$

Determine the phase angle at $\omega = 0.1\ \text{rad/s}$.

$$\phi = -90° + \tan^{-1}(0.1) - 2\tan^{-1}\left(\frac{0.1}{5}\right)$$
$$= -86.58°$$

Determine the phase angle at $\omega = 1\ \text{rad/s}$.

$$\phi = -90° + \tan^{-1}(1) - 2\tan^{-1}\left(\frac{1}{5}\right)$$
$$= -67.62°$$

Determine the phase angle at $\omega = 10\ \text{rad/s}$.

$$\phi = -90° + \tan^{-1}(10) - 2\tan^{-1}\left(\frac{10}{5}\right)$$
$$= -132.58°$$

Determine the phase angle at $\omega = 100\ \text{rad/s}$.

$$\phi = -90° + \tan^{-1}(100) - 2\tan^{-1}\left(\frac{100}{5}\right)$$
$$= -174.85°$$

Determine the phase angle at $\omega = 1000\ \text{rad/s}$.

$$\phi = -90° + \tan^{-1}(1000) - 2\tan^{-1}\left(\frac{1000}{5}\right)$$
$$= -179.48°$$

Step 7 of 9

Draw the asymptotic phase plot.

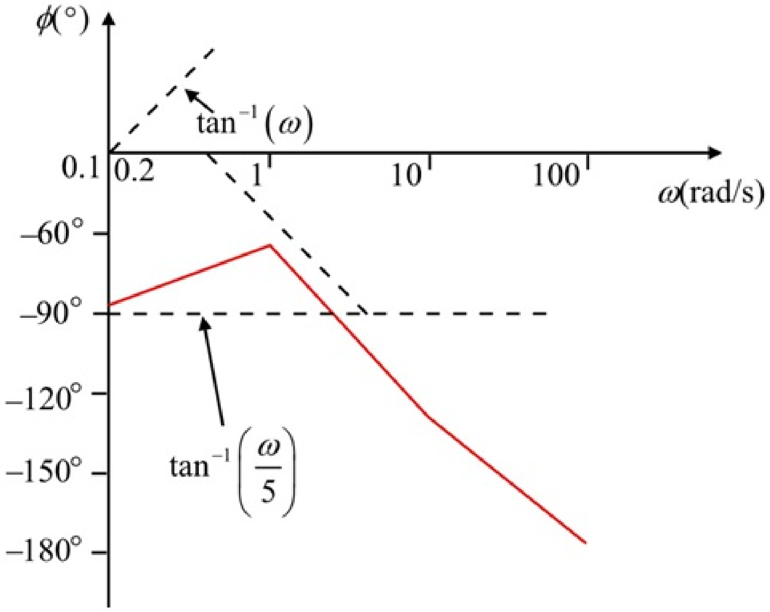

Figure 2

Step 8 of 9

MATLAB code to plot Bode diagram:

```
>> num=[250 250];
>> den=[1 10 25 0];
>> x=tf(num,den);
>> bode(x)
>> grid
```

Step 9 of 9

The Bode diagram is shown in Figure 3.

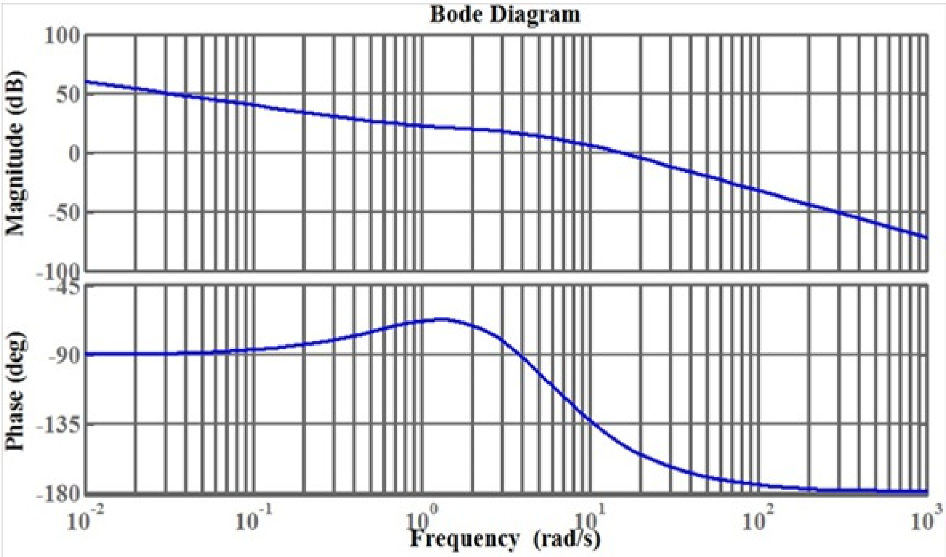

Figure 3

Thus, the Bode plots are drawn.